

Flow Matching II

Discussion #6

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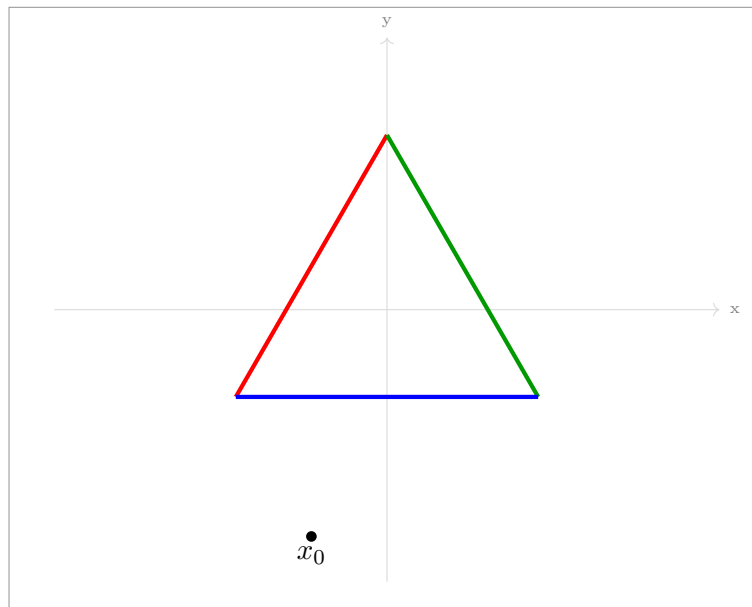
Logistics

Some key announcements:

- The midterm is **10/15**. You are allowed one handwritten cheat sheet (front and back).
- Project 3 is due **10/20**!

Warmup: Guidance

Now suppose we want to draw from a distribution corresponding to sides of a triangle (think of this like our little image manifold). In addition, we want to *control* what type of point our model generates: **red**, **green**, or **blue**. (not the color, but the location)



0.1 Adding control. The most common way to control diffusion models is by additional *conditioning* input, like language, to the model.

Explain how one could sample from a specific color without additional model input by changing training.

Train one model per color. We will have: Red model, Green model, and Blue model.

Classifier-Free Guidance (CFG) modifies the vector field to nudge samples more towards red compared to an unconditional sample:

$$\tilde{u}_\theta(x, t, c) = u_\theta(x, t, \varnothing) + w \cdot (u_\theta(x, t, y) - u_\theta(x, t, \varnothing))$$

where u_θ is our neural network model, which learns marginal flow.

0.2 Prompted Flow: Draw in the diagram where $u_\theta(x, t, \text{red})$ and $u_\theta(x, t, \text{green})$ point.

Each will point towards the center of the red and green lines respectively (because of the same effect as 1.5. This is because we're at full noise!)

0.3 Unprompted Flow: Draw in the diagram where $v(x_0, t, \varnothing)$ points

It will point to the center of the triangle (center of *all* the data).

0.4 Guidance: Draw the CFG vector $\tilde{u}_\theta$ for a large w . How does it differ from the standard prompted vector?

With a high w , it will point more and more away from the center of the triangle and more towards the side of the triangle.

1 Flow Matching Implementation

In last week's discussion, we explored geometric intuition for conditional and marginal flows. This week, we'll connect this idea to the actual implementation.

Consider a toy 1D flow matching problem. Our “dataset” has just two points: $A = +0.5$ and $B = -0.5$, each sampled with probability 0.5. We run a simple flow matching training loop:

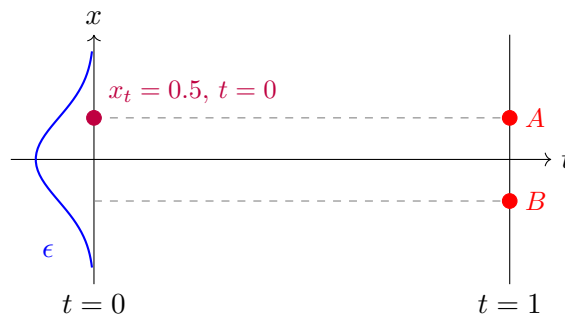
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L1: Sample  $x_1 \sim \text{data}$  (A or B with prob 0.5 each)
L2: Sample  $\epsilon \sim \mathcal{N}(0, 1)$ 
L3: Sample  $t \sim \text{Uniform}(0, 1)$ 
L4: Compute  $x_t = (1 - t)\epsilon + t \cdot x_1$ 
L5: Compute target velocity:  $u = x_1 - \epsilon$ 
L6: Update  $\theta$  to minimize  $\|u_\theta(x_t, t) - u\|^2$ 

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For a given (x_t, t) , the training loop sometimes supervises toward A and sometimes toward B .

1.1 Consider the case where $t = 0$ is sampled at L3, and $x_t = 0.5$ is computed at L4.



(a) If $x_1 = A$ was sampled at L1, what ϵ was sampled at L2? What if $x_1 = B$?

Both require $\epsilon = 0.5$. At $t = 0$, L4 gives $x_t = \epsilon$ regardless of x_1 .

(b) Using L5, compute the target velocity u for each case. ($x_1 = A$ and $x_1 = B$).

For $x_1 = A$: $u = A - 0.5 = 0.5 - 0.5 = 0$
 For $x_1 = B$: $u = B - 0.5 = -0.5 - 0.5 = -1$

(c) How often does the training loop produce $(x_t=0.5, t=0, x_1=A)$ vs. $(x_t=0.5, t=0, x_1=B)$?

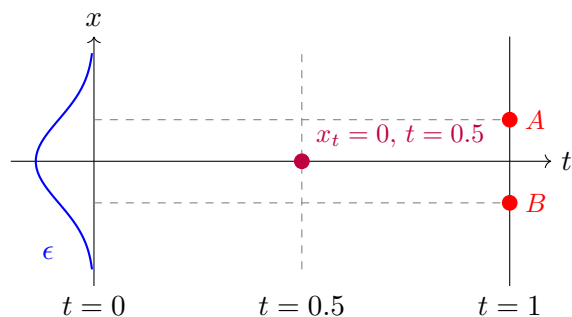
Equally often—both need the same ϵ , and L1 picks A vs B with 50/50 probability.

(d) What does $u_\theta(0.5, 0)$ converge to after training? Which x_1 does this velocity vector point at?

$$u_\theta(0.5, 0) = \frac{1}{2}(0) + \frac{1}{2}(-1) = -0.5.$$

Starting from $x_0 = 0.5$, this velocity vector points at $x_1 = 0.5 + (-0.5) = 0$.

1.2 Consider the case where $t = 0.5$ is sampled at L3, and $x_t = 0$ is computed at L4.



(a) If $x_1 = A$ was sampled at L1, what ϵ was sampled at L2? What if $x_1 = B$?

For A : $\epsilon = -0.5$. For B : $\epsilon = +0.5$. (Both equally likely under $\mathcal{N}(0, 1)$.)

(b) Using L5, compute the target velocity u for each case.

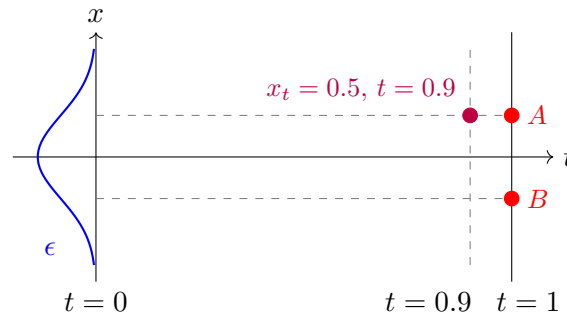
$$\text{For } x_1 = A: u = A - (-0.5) = 0.5 + 0.5 = 1$$

$$\text{For } x_1 = B: u = B - 0.5 = -0.5 - 0.5 = -1$$

(c) What does $u_\theta(0, 0.5)$ converge to?

$$u_\theta(0, 0.5) = \frac{1}{2}(1) + \frac{1}{2}(-1) = 0$$

1.3 Consider the case where $t = 0.9$ is sampled at L3, and $x_t = 0.5$ is computed at L4.



(a) If $x_1 = A$ was sampled at L1, what ϵ was sampled at L2? What if $x_1 = B$?

For A : $\epsilon = 0.5$ (common). For B : $\epsilon = 9.5$ (essentially never sampled).

(b) Using L5, compute the target velocity u for each case.

For $x_1 = A$: $u = A - 0.5 = 0.5 - 0.5 = 0$
 For $x_1 = B$: $u = B - 9.5 = -0.5 - 9.5 = -10$

(c) How often does the training loop produce supervision toward A vs. toward B at this (x_t, t) ?

Almost always A . The B case requires $\epsilon = 9.5$, which is > 9 standard deviations from the mean.

(d) What does $u_\theta(0.5, 0.9)$ converge to?

$u_\theta(0.5, 0.9) \approx 0$ (the A case dominates)

1.4 Both 1.1 and 1.3 have $x_t = 0.5$. Why does u_θ converge to -0.5 at $t = 0$ but ≈ 0 at $t = 0.9$?

At $t = 0$, L4 computes $x_t = \epsilon$ regardless of x_1 , so both targets appear equally often.
 At $t = 0.9$, reaching $x_t = 0.5$ with target B requires an implausible ϵ at L2, so target A dominates.
 The network learns a weighted average of target velocities. The weights depend on how likely each (x_t, t, x_1) combination is under the training distribution.

2 Mixing and Matching Models and Samplers

Suppose you have a pre-trained **ϵ -prediction model** $\epsilon_\theta(x_t, t)$ that predicts noise (like DDPM), and a sampler that implements **Euler integration** for flow matching:

$$x_{t+\Delta t} = x_t + \Delta t \cdot u_\theta(x_t, t)$$

The sampler expects a velocity u , but your model outputs ϵ .

2.1 Using the flow matching interpolation $x_t = (1-t)\epsilon + t \cdot x_{\text{clean}}$ and velocity definition $u = x_{\text{clean}} - \epsilon$, derive a formula for u in terms of x_t , ϵ , and t .

From the two equations:

$$\begin{aligned} x_t &= (1-t)\epsilon + t \cdot x_{\text{clean}} \\ u &= x_{\text{clean}} - \epsilon \end{aligned}$$

From the second equation: $x_{\text{clean}} = u + \epsilon$.

Substituting into the first:

$$\begin{aligned} x_t &= (1-t)\epsilon + t(u + \epsilon) \\ x_t &= (1-t)\epsilon + tu + t\epsilon \\ x_t &= \epsilon + tu \end{aligned}$$

Therefore: $\epsilon = x_t - tu$, which gives us $u = \frac{x_t - \epsilon}{t}$

2.2 Rewrite the Euler update step using ϵ_θ instead of u_θ .

Substituting $u = \frac{x_t - \epsilon}{t}$:

$$x_{t+\Delta t} = x_t + \Delta t \cdot \frac{x_t - \epsilon_\theta(x_t, t)}{t}$$